

Math 403 Homework #2, Spring 2025

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Collaborators:

(1 point for each of up to 3 collaborators who also list you)

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Due: 11:59pm Saturday 8 February 2025

READING ASSIGNMENTS

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7. for Thu. 30 January
 - [Serge Lang, *Linear Algebra*, Chapter XII, §1–§2] separating hyperplanes
 - [Serge Lang, *Linear Algebra*, Chapter XII, §3–§4] support hyperplane, extreme point
8. for Tue. 4 February
 - [Wikipedia, *Grassmannian*] as much as you're willing; thru §3 (“*as a set*”) at least
 - [Wikipedia, *Group action*] get a feel for the concept; no need to overdo it
9. for Thu. 6 February
 - [multivariable calculus, any source]: open & closed sets; differentiability; chain rule $D(f \circ g) = Df \circ Dg$; inverse function theorem: Df invertible $\Rightarrow f$ locally invertible
 - [Wikipedia, *Topological space*] get a feel; through §5.1 (*Metric spaces*), say
 - [Wikipedia, *Differentiable manifold*] read up through §2 (*Definition*)
10. for Tue. 11 February
 - [Lax, 7.Isometry, p.87–89] isometry, orthogonal group
 - [Lax, 7.Complex Euclidean structure, p.95–96] unitary matrices
 - [Treil, §5.6–§5.7] unitary transformations, rigid motions
11. for Thu. 13 February
 - [Treil, §6.1–§6.2] Spectral theorem, normal operators

EXERCISES

1. Find the Jordan form of a 5×5 matrix A whose sole eigenvalue is 3, given that $A - 3I$, $(A - 3I)^2$, $(A - 3I)^3$, and $(A - 3I)^4$ have kernels of dimension 2, 3, 4, and 5, respectively. Do the same thing if the kernel dimensions are 2, 4, 5, and 5, respectively.

Solution

$$\begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix} \quad (1)$$

Upto permutations of jordan blocks.

$$\begin{bmatrix} 3 & 1 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix} \quad (2)$$

Upto permutations of jordan blocks.

2. Find a Jordan form and a corresponding Jordan basis for the matrix

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$$A = \begin{bmatrix} 7 & 1 & 2 & 2 \\ 1 & 4 & -1 & -1 \\ -2 & 1 & 5 & -1 \\ 1 & 1 & 2 & 8 \end{bmatrix}.$$

Solution

The characteristic polynomial for this matrix is $p(x) = (x - 6)^4$. The geometric multiplicity is 2. Which means that there are 2 jordan blocks. The eigen vectors are.

$$v_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$
$$v_2 = \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

The geometric multiplicities are 2,3 and 4 for $(A - 6I)$, $(A - 6I)^2$, and $(A - 6I)^3$ respectively. So the matrix's jordan form is as below.

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 1 & 0 \\ 0 & 0 & 6 & 1 \\ 0 & 0 & 0 & 6 \end{bmatrix} \quad (3)$$

3. Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$ of finite dimension. Let $\varphi : V \hookrightarrow W$ be an injective homomorphism. Show that if μ is a norm on W then $\mu \circ \varphi$ is a norm on V .

Solution

For $\mu \circ \varphi$ to be a norm on W it has to satisfy certain properties of a norm.

1. $\mu \circ \varphi(x) = 0 \Leftrightarrow x = 0$

Since μ is already a norm, $\mu(x) = 0 \Leftrightarrow x = 0$, but φ is injective therefore $\ker(\varphi) = 0$, so $\mu \circ \varphi(x) = 0 \Leftrightarrow x = 0$.

2. $\mu \circ \varphi(\alpha x) = |\alpha| \mu \circ \varphi(x)$ for $\alpha \in$ the field.

Since μ is a norm, then $\mu(\alpha x) = |\alpha| \mu(x)$, and since φ is linear $\varphi(\alpha x) = \alpha \varphi(x)$. $\mu \circ \varphi(\alpha x) = \mu(\alpha \varphi(x)) = |\alpha| \mu \circ \varphi(x)$, satisfying the property.

3. $\mu \circ \varphi(x + y) \leq \mu \circ \varphi(x) + \mu \circ \varphi(y)$ Since μ is a norm $\mu(x + y) \leq \mu(x) + \mu(y)$. So $\mu \circ \varphi(x + y) = \mu(\varphi(x) + \varphi(y)) \leq \mu \circ \varphi(x) + \mu \circ \varphi(y)$.

So $\mu \circ \varphi$ is a norm on W .

4. Let ν be a norm on $V = \mathbb{R}^n$ or $\mathbb{C}^n$ with unit ν -ball $B_\nu = \{\mathbf{x} \in V \mid \nu(\mathbf{x}) \leq 1\}$. Prove [/3](#) that B_ν is closed, bounded, convex, and *equilibrated*: $\mathbf{x} \in B_\nu$ and $|\alpha| \leq 1 \Rightarrow \alpha\mathbf{x} \in B_\nu$. Also show that the origin lies interior to B_ν . Conversely, if $B \subseteq V$ is closed, bounded, convex, equilibrated subset with 0 in its interior, then a norm $\nu_B : V \rightarrow \mathbb{R}$ with unit ball B can be defined by $\nu_B(\mathbf{x}) = \inf\{\alpha > 0 \mid \frac{1}{\alpha}\mathbf{x} \in B\}$.

Solution

Claim 1: B_ν is closed, bounded, convex and equilibrated.

Closed:- Assume for the sake of contradiction, that there exist a sequence of points inside B_ν such that it converges to a value a , such that $a \notin B_\nu$. Therefore $\nu(a) > 1$, but since the a is a the limit of some sequence all of whose elements are in B_ν then $\nu(a) \leq 1$, which contradicts the assumption that $a \notin B_\nu$.

Bounded:- Take $v, s \in B_\nu$. $\nu(V) \leq 1, \nu(S) \leq 1 \Rightarrow \nu(V + S) \leq \nu(V) + \nu(S)$ So B_ν is bounded with an upper bound of 2.

Convex:- Take any two points, $x, y \in B_\nu$.

All points $\alpha x + (1 - \alpha)y, \alpha \in [0, 1]$ are in B_ν .

Proof:- Since $\nu(\alpha x + (1 - \alpha)y) \leq \alpha\nu(x) + (1 - \alpha)\nu(y)$

$\nu(x) \leq 1$ and $\nu(y) \leq 1$, so $\alpha\nu(x) + (1 - \alpha)\nu(y) \leq \alpha + (1 - \alpha) = 1$.

So all points have norm ≤ 1 , and are therefore in the set. So B_ν is convex.

Equilibrated: For $|x| \leq 1, \alpha x \in B_\nu$.

$\nu(x) \leq 1, \nu(\alpha x) = |\alpha|\nu(x) \leq |\alpha| \leq 1 \Rightarrow \alpha x \in B_\nu$.

Origin lies in the interior: Take the open ball of radius 1/2. This ball is contained inside B_ν , and since it is open all the points are in the interior.

Converse claim:- If B is a closed, bounded, convex, equilibrated subset with 0 in its interior then, a norm $\nu_B : V \rightarrow \mathbb{R}^+$ with unit ball B can be defined by $\nu_B(x) = \inf\{\alpha \geq 0 \mid 1/\alpha \in B\}$

Property #1: $\nu_B(x) = 0 \Leftrightarrow x = 0$

This is trivially true(atleast by setting $\nu_B(0) = 0$ in the definition).

Property #2: $\nu(\beta x) = |\beta|\nu(x)$

Lemma :- $\nu(x) = \nu(-x)$

Proof:- By equilibration, if $x \in B$, then $-x \in B$ as well.

Let $\nu(x) = \alpha_1$ and $\nu(-x) = \alpha_2$. If $\alpha_1 \neq \alpha_2$ then without loss of generality (WLOG) assume $\alpha_1 > \alpha_2$.

$1/\alpha_2 x \in B$, but this also implies $\frac{-1}{\alpha_2} x \in B$, so $\frac{1}{\alpha_2} x \in B$, but $\alpha_2 < \alpha_1$ and therefore contradiction, therefore $\alpha_1 = \alpha_2$.

Assume $\nu(\beta x) = \alpha \Leftrightarrow \frac{\beta}{\alpha} x \in B \Leftrightarrow \frac{\beta}{-\alpha} \in B \Rightarrow \frac{\beta}{|\alpha|} \in B$.

Property # 3: $\nu(x + y) \leq \nu(x) + \nu(y)$

$$\left(\frac{1}{\alpha_1 + \alpha_2}\right)(x + y) = \left(\frac{1}{\alpha_1 + \alpha_2 - 6 - 697}\right)$$

5. The Hahn–Banach theorem says that for any norm ν on $\mathbb{C}^n$ and any subspace $V \subseteq \mathbb{C}^n$, /3
if $\varphi : V \rightarrow \mathbb{C}$ is a linear functional whose maximum magnitude $|\varphi(\mathbf{x})|$ for $\mathbf{x}$ in the unit ν -sphere of V is α , then φ can be extended to a linear functional on $\mathbb{C}^n$ whose maximum magnitude on the unit ν -sphere in $\mathbb{C}^n$ is α . Show that the Hahn–Banach theorem is equivalent to the statement that separating hyperplanes exist for points outside of any closed, bounded, convex, equilibrated set that contains 0 in its interior.
6. Let V be the vector space of real (or complex) absolutely convergent series. The /3
function $x \mapsto \|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{\frac{1}{p}}$ for $p \geq 1$ is a norm on V . (You may assume this, although no complaints if you'd like to prove it.) Prove that $\|x\|_p$ is not topologically equivalent to $\|x\|_q$ if $p < q$.
7. Sketch the convex hull of $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ and $\left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. /3
8. Fix an operator $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$. If $X \subseteq \mathbb{R}^n$ is convex and $\mathbf{x}$ is an extreme point of X , /3
must $\varphi(\mathbf{x})$ be an extreme point of $\varphi(X)$? What if φ is assumed invertible?

Solution

For the first part, the answer is no.

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

You can see that the matrix is singular and collapses the space to a line, so therefore the extreme points are no longer extreme points after the transformation. For example, the square with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$, the extreme points are $(0, 0)$, $(1, 1)$, $(0, 1)$, $(1, 0)$ but after the transformation the extreme points are $(0, 0)$ and $(0, 1)$.

For the second part, the answer is yes, because the transformation is invertible, therefore any line in the image can be mapped back to a line in the original space, if a point is an extreme point in X , and not extreme in the image, then it must be a convex combination of points in the output space, but since the transformation is invertible, you can take the inverse of the line to show that the point itself couldn't have been an extreme point, contradicting the assumption.

9. Prove that the intersection of finitely many closed halfspaces in $\mathbb{R}^n$ can have only /3 finitely many extreme points. Must it have any extreme points at all? If it does have extreme points, must the intersection equal the convex hull of its extreme points?

Solution

Claim #1: Extreme points can only be on the boundary points of the intersection.

Proof: Assume for the sake of contradiction that there exists an extreme point x that is not on the boundary of the intersection. Then x has an open ball around it that is contained in the intersection. Then you can take two points in the open ball, and take a convex combination of the two points, which contains x , contradicting the assumption that x is an extreme point.

Claim #2: There can be at most 2 extreme points on the boundary of the halfspaces.

If there are more than 2, then you can take a convex combination of the two points, which contains all the other points, contradicting the assumption that the other points are extreme points.

Conclusion: Since there are finitely many halfspaces, and each halfspace has a finite number of extreme points, the intersection has a finite number of extreme points.

10. Fix a column vector $\mathbf{b} \in \mathbb{R}^n$ and an $n \times n$ real matrix A . Show that the set of solutions [/3](#)
 $\{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} \leq \mathbf{b}\}$ of the inequality $A\mathbf{x} \leq \mathbf{b}$ is convex, where $\mathbf{a} \leq \mathbf{b}$ if $a_i \leq b_i$ for all i .
What kind of convex set is it? (No need to provide full justification of this last bit.)

Solution

Since you can write the inequality of each entry as a halfspace, and halfspaces are convex, then the intersection of the halfspaces is convex. The set is a polyhedron.

11. A *cone* is a subset $C \subseteq \mathbb{R}^n$ closed under nonnegative scaling: $\mathbf{x} \in C$ implies $\alpha\mathbf{x} \in C$ for all $\alpha \in \mathbb{R}_{\geq 0}$. A ray $\mathbb{R}_{\geq 0}\mathbf{x} = \{\alpha\mathbf{x} \mid \alpha \in \mathbb{R}_{\geq 0}\} \subseteq C$ is *extreme* if $\mathbf{x} \in \mathbf{y} + C \subseteq C$ implies $\mathbf{y} \in \mathbb{R}_{\geq 0}\mathbf{x}$. Let ν be an outer support vector for a closed convex cone C , so $\langle \nu, \mathbf{x} \rangle \leq 0$ for all $\mathbf{x} \in C$. Assume ν attains a unique maximum on C . Prove that the minimum angle between ν and a unit vector $\mathbf{x} \in C$ occurs when $\mathbf{x}$ lies on an extreme ray of C . /3

Solution

12. How many $k \times n$ matrices of rank k with entries in a field $\mathbb{F}_q$ of size q are there? /3
Hint: how many nonzero row vectors are there in $\mathbb{F}_q^n$? How many vectors are linearly independent from the first one? How many are independent from the first two?
13. Show that the action of $GL_k(F)$ on the set of $k \times n$ matrices of rank k is *free*: if $g \neq g'$ are invertible $k \times k$ matrices and A is a $k \times n$ matrix of rank k , then $gA \neq g'A$. Use this (and the previous exercise) to compute the cardinality of the set $G_k(\mathbb{F}_q^n)$ of k -planes in $\mathbb{F}_q^n$. Hint: use #12 twice, once with $k = n$ and once with $k < n$. /3
14. Does the first claim in #13 remain true if the rank of A is less than k ? /3
15. Use #13 to count the number of Sets in the game of Set. (Note: a Set is an affine line, not a linear subspace of dimension 1, so this is not quite a special case of #13. There is more than one way around this; I'll leave it to you to find at least one.) Confirm your count by counting a different way: a line L is determined by any pair of points on L . /3